

Which markets deserve deeper energy diligence?

A decision screen for corporate energy and site-selection teams across 50 non-random markets.

Insights and decision use | Five minutes



Data scope and general observations

50

non-random markets selected for breadth and data availability

47/50

household price records (94% coverage)

45/50

business price records (90% coverage)

47/50

records for consumption, mix, and balance (94%)

Reference periods differ: prices use 2023-2026 average; consumption, production, and electricity mix use 2023.

Observed ranges: household price 0.006-0.414 USD/kWh; non-fossil share 0.9-100.0%; consumption 0.231-169.459 EJ; balance gap -31.0 to +35.5 EJ.

Complete-case correlations range from $r=-0.138$ to 0.425 with $n=43-45$. They describe association only.

EVIDENCE TABLE

Detailed comparison

Sort by Electricity price

Download CSV

MARKET	PRICE	CONSUMPTION	NON-FOSSIL	BALANCE / TRADE	PERIOD
Denmark	0.355 USD/kWh	0.7 EJ	86.4 %	+0.3 EJ	2023-2026 average 2023
Switzerland	0.367 USD/kWh	1.1 EJ	98.0 %	+0.7 EJ	2023-2026 average 2023
Germany	0.406 USD/kWh	11.4 EJ	56.1 %	+8.2 EJ	2023-2026 average 2023
Belgium	0.406 USD/kWh	2.3 EJ	73.5 %	+1.9 EJ	2023-2026 average 2023
Italy	0.414 USD/kWh	6.1 EJ	44.6 %	+5.2 EJ	2023-2026 average 2023

Price and generation contrasts

European price peak

Italy leads household and business prices at \$0.414 and \$0.411 per kWh. Germany and Belgium each report \$0.406 household prices.

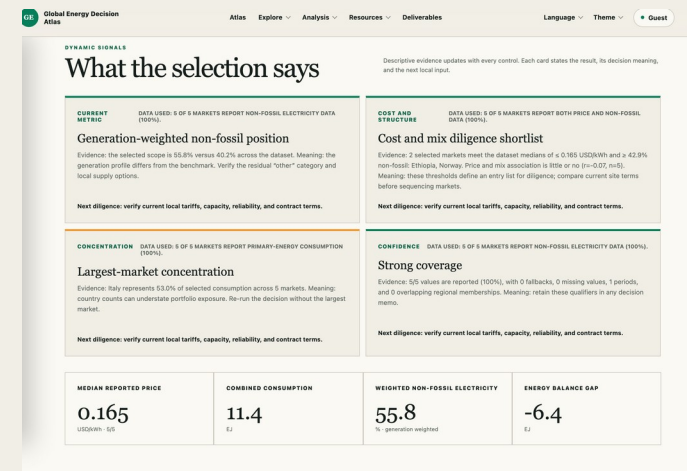
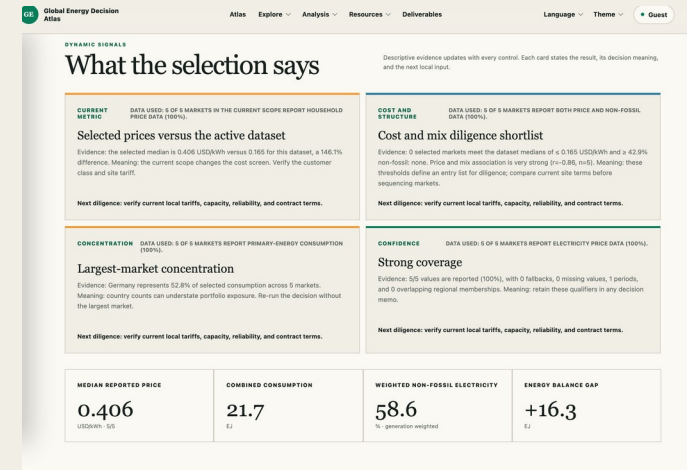
Low price does not imply the same generation structure

Ethiopia: \$0.006, 100.0% non-fossil

Norway: \$0.165, 98.5% non-fossil

Algeria: \$0.041, 0.9% non-fossil

Tariff structure, contracts, and local grid conditions require separate checks.



Consumption concentration

31.1%

China

17.3%

United States

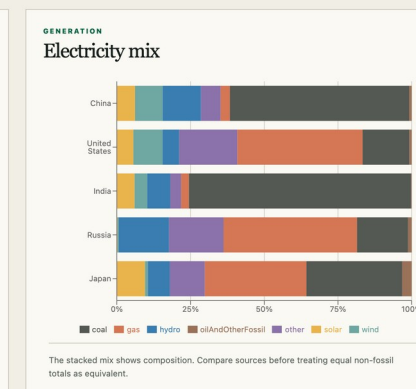
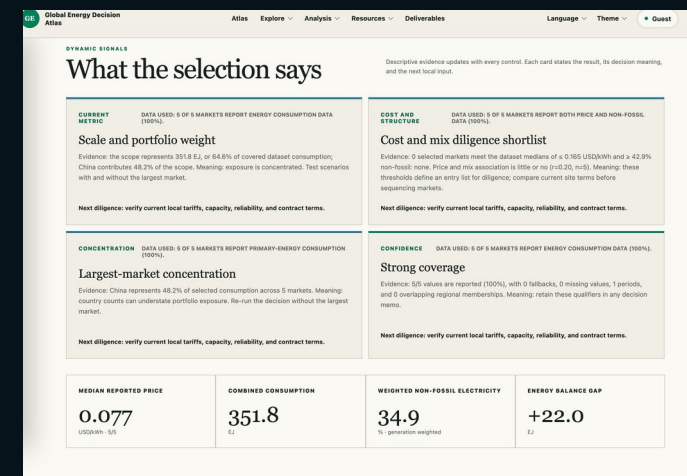
48.4%

China plus United States

64.6%

top five markets

Shares use 545.0 EJ reported across complete records. A portfolio with equal country counts can still have a highly concentrated energy footprint.



Energy balance positions

Primary energy consumption minus total energy production

+35.5 EJ

China

+16.2 EJ

India

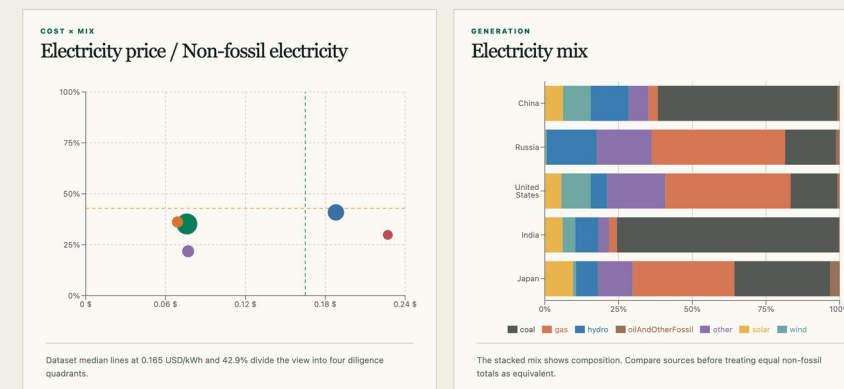
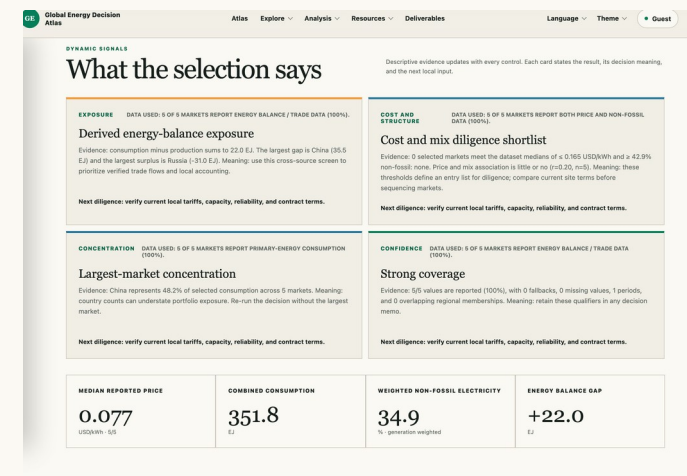
-31.0 EJ

Russia

-14.0 EJ

United States

Positive values indicate a consumption gap.
Negative values indicate production above consumption. The measure does not observe actual trade flows.



Decision use and diligence shortlist

China, India, and United States

Start with scale and balance exposure.

Ethiopia, Norway, Italy, and Algeria

Test the price and generation contrasts against local operating conditions.

Local checks before a decision

Facility tariff and demand charges
Contract term, indexation, and hedging
Available grid capacity and connection cost
Reliability, curtailment, and outage history

GUIDED ANALYSIS

Four views that start with a decision question

Each view applies a fixed, reproducible configuration. The Atlas evidence comes from this snapshot; the linked sources provide market context.

01 Europe's price high point Active

ATLAS EVIDENCE
Italy records the highest household and business prices in Expanded 50 at 0.414 and 0.411 USD/kWh. Germany and Belgium both record 0.406 USD/kWh for households.

MARKET CONTEXT
The pattern may reflect differences in wholesale costs, taxes, levies, and network charges. The linked IEA and Eurostat material explains why a retail price needs a component-level review.

Management meaning: Treat the price ranking as a prompt to request the applicable industrial tariff, demand charges, and contract terms before comparing sites.

[IEA Italy 2023](#) [Eurostat electricity price data](#)

[View active](#)

02 Ethiopia's price and structure contrast

ATLAS EVIDENCE
Ethiopia combines the lowest household price, 0.006 USD/kWh, with 100.0% non-fossil generation and 98.4% hydro. Algeria records 0.9% non-fossil generation in the same comparison.

MARKET CONTEXT
The contrast may reflect very different power systems, access conditions, and tariff structures. The World Bank compact provides context for Ethiopia's expansion priorities and financing needs.

Management meaning: Low national price and a high non-fossil share do not settle site viability. Ask about connection capacity, reliability, customer class, and supply terms.

[World Bank Ethiopia National Energy Compact](#)

[Open this view](#)

03 Consumption scale in China and the United States

ATLAS EVIDENCE
China accounts for 31.1% and the United States 17.3% of the 545.0 EJ reported total. Together they represent 48.4%; the five largest markets represent 64.6%.

MARKET CONTEXT
This concentration may reflect industrial scale, population, and energy demand across sectors. IEA Electricity 2025 supplies current context for China's power demand and supply outlook.

Management meaning: A country count can hide portfolio concentration. Test recommendations with and without the two largest markets.

[IEA Electricity 2025](#)

[Open this view](#)

04 Diverging energy balance positions

ATLAS EVIDENCE
The screening proxy is positive for China (35.5 EJ) and India (16.2 EJ), and negative for Russia (-31.0 EJ) and the United States (-14.0 EJ). Positive means consumption exceeds production in this cross-source calculation.

MARKET CONTEXT
The split may reflect domestic production, demand, and accounting differences. EIA material on the United States and Russia helps interpret the national positions without relabeling the proxy as observed trade.

Management meaning: Use the balance gap to prioritize diligence on verified trade flows, fuel exposure, and local reliability, not as a final import measure.

[EIA United States energy balance](#) [EIA Russia country analysis](#)

[Open this view](#)

This sequence orders investigation. It does not rank the best market.

Method, evidence boundaries, and limitations

Analysis status

Descriptive / performed

Diagnostic / partial

Predictive / not performed

Prescriptive / not performed

Limits that affect interpretation

Missing values change denominators.

Reference years differ. National averages do not represent a facility contract or connection point. Complete-case correlations do not establish causation.

11 stable files | global-energy-decision-atlas.vercel.app/deliverables

COURSE FILES

Deliverables

Download the evidence, method, demonstration, and reflection used to support this site.

01

The collected dataset in CSV format

Original Assignment 15 data, Expanded 50 snapshot, and the source manifest.

Assignment 15 CSV

Open Download

CSV

Expanded 50 CSV

Open Download

CSV

Source manifest

Open Download

JSON

02

A one-page data and methodology note

Publication-ready PDF and editable DOCX.

Methodology PDF

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PDF

Editable methodology

Download

DOCX

03

A five-minute insights and decision-use presentation

The formal seven-slide presentation plus a supplemental site design walkthrough, each in PPTX and PDF.

Insights and decision use presentation

Download

PPTX

Insights and decision use presentation

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PDF

Supplemental site design walkthrough

Download

PPTX

Supplemental site design walkthrough

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PDF

04

A reflection on the findings and limits

A three-page PDF and Markdown source with four data stories, analysis boundaries, a diligence shortlist, and the next data request.

Reflection PDF

Open Download

PDF

Reflection Markdown

Open Download

Markdown